

ROQUEMAURE, QC, Canada

WMO: 717361

Lat: 48.630N

Lon: 79.460W

Elev: 269

StdP: 98.13

Time Zone: -5.00 (NAE)

Period: 93-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.6	-33.2	-41.4	0.1	-36.4	-37.9	0.1	-32.9	8.8	-11.1	8.2	-12.7	1.2	180	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	28.5	20.5	26.7	19.4	25.0	18.2	21.8	26.3	20.7	24.8	19.6	23.3	3.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.3	15.5	24.1	19.2	14.5	22.9	18.1	13.5	22.1	65.1	26.6	61.0	24.5	57.1	23.2	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.4	6.5	-41.6	31.4	2.7	1.9	-43.6	32.8	-45.1	33.9	-46.7	35.0	-48.6	36.4		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.4	-18.0	-15.8	-8.2	-0.1	8.4	15.1	17.0	15.6	11.3	4.5	-3.4	-11.0	(6)
(7)		DBStd	13.47	8.43	8.18	8.00	5.66	4.71	4.67	3.36	3.67	4.52	4.56	6.04	8.34	(7)
(8)		HDD10.0	3829	868	723	566	304	90	9	0	2	35	181	402	649	(8)
(9)		HDD18.3	6258	1127	957	824	552	310	118	66	102	217	427	651	908	(9)
(10)		CDD10.0	684	0	0	0	2	40	163	218	176	73	12	0	0	(10)
(11)		CDD18.3	71	0	0	0	0	1	22	26	17	5	0	0	0	(11)
(12)		CDH23.3	753	0	0	0	0	43	258	236	162	55	0	0	0	(12)
(13)		CDH26.7	165	0	0	0	0	6	66	49	30	14	0	0	0	(13)
(14)	Wind	WSAvg	2.9	2.7	3.0	2.9	2.8	3.1	2.6	2.6	2.5	2.8	3.4	3.2	2.9	(14)
(15)	Precipitation	PrecAvg	799	50	36	43	50	66	74	96	86	95	83	66	54	(15)
(16)		PrecMax	985	114	76	82	100	121	129	178	134	183	143	125	93	(16)
(17)		PrecMin	657	16	6	6	17	24	44	47	29	40	34	37	29	(17)
(18)		PrecStd	75	23	18	16	21	25	25	31	26	29	28	21	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	6.7	12.0	18.4	27.7	31.2	30.9	29.7	28.6	20.8	13.8	4.5	(19)
(20)			MCWB	2.6	5.1	7.1	13.0	17.6	21.8	22.9	21.1	19.2	15.9	10.6	2.3	(20)
(21)		2%	DB	0.1	3.3	8.3	14.5	24.1	28.5	27.7	27.1	24.3	17.4	10.3	2.7	(21)
(22)			MCWB	-0.4	1.4	5.1	8.3	14.4	20.1	20.8	19.8	18.6	13.8	9.4	1.9	(22)
(23)		5%	DB	-2.1	0.5	5.5	11.3	21.0	26.4	25.9	25.2	21.2	14.6	7.2	1.7	(23)
(24)			MCWB	-2.6	-0.3	2.6	5.7	12.9	18.4	19.6	19.2	17.1	12.3	5.8	1.0	(24)
(25)		10%	DB	-4.5	-2.2	3.2	8.8	18.0	24.1	24.1	23.0	19.3	12.1	4.3	0.5	(25)
(26)			MCWB	-5.1	-3.3	1.0	4.7	10.7	17.2	18.5	18.1	15.9	10.0	3.1	-0.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	5.8	7.8	13.5	17.2	23.1	24.0	22.6	20.6	16.7	11.0	3.1	(27)
(28)			MCDWB	2.9	6.7	11.4	17.2	24.4	29.2	29.1	26.5	24.6	19.4	12.1	3.6	(28)
(29)		2%	WB	-0.6	1.4	5.1	9.3	14.9	21.3	22.1	21.2	19.4	14.8	9.1	2.0	(29)
(30)			MCDWB	-0.3	2.9	7.7	11.0	19.8	27.0	25.8	25.4	23.1	17.0	10.6	2.5	(30)
(31)		5%	WB	-3.0	-0.5	2.6	6.7	13.1	20.1	20.7	20.0	18.0	12.6	6.3	1.1	(31)
(32)			MCDWB	-2.3	0.5	4.6	10.1	17.7	24.3	24.0	24.1	21.4	14.5	7.6	1.7	(32)
(33)		10%	WB	-5.4	-3.8	1.0	4.7	11.6	18.6	19.2	18.8	16.0	10.2	3.4	-0.1	(33)
(34)			MCDWB	-4.7	-2.9	2.5	8.2	15.8	22.4	22.6	22.3	18.3	12.0	4.5	0.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.7	15.1	13.5	12.1	12.8	13.0	11.1	11.8	10.8	8.3	7.7	9.7	(35)
(36)			MCDBR	11.8	12.8	13.0	15.9	18.0	15.7	13.9	14.3	14.0	11.8	9.5	6.3	(36)
(37)			MCWBR	11.2	10.8	10.1	9.3	10.1	8.9	7.8	8.2	8.7	8.7	8.2	5.9	(37)
(38)		5% WB	MCDWB	11.8	12.0	11.4	12.4	14.5	13.0	12.1	12.0	12.8	10.6	9.0	5.8	(38)
(39)			MCWBR	11.8	10.9	9.0	9.2	8.7	8.4	7.4	7.5	8.5	8.3	7.9	5.7	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.261	0.278	0.311	0.351	0.359	0.377	0.393	0.381	0.364	0.320	0.293	0.255	(41)	
(42)		taud	2.283	2.241	2.220	2.293	2.398	2.370	2.360	2.386	2.417	2.529	2.460	2.351	(42)	
(43)		Ebn at Noon	809	876	900	897	902	884	862	858	837	829	773	773	(43)	
		Edh at Noon	71	95	116	120	113	117	117	109	96	73	62	59		
(44)	All-Sky Solar Radiation	RadAvg	1.24	2.34	3.74	4.61	5.19	5.68	5.38	4.51	3.37	1.97	1.02	0.82	(44)	
(45)		RadStd	0.20	0.33	0.37	0.38	0.48	0.35	0.47	0.30	0.30	0.27	0.12	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page